

Trigonometry

This section covers questions on trigonometric identities, compound angles, and solving trigonometric equations.

1. Which of the following is correct? (a) $1 - \tan^2\theta = \sec^2\theta$ (b) $1 + \tan^2\theta = \sec^2\theta$ (c) $1 - \cot^2\theta = \sec^2\theta$ (d) $1 + \cot^2\theta = \sec^2\theta$ (e) None of the above

- **Solution:** The Pythagorean identity is $\sin^2\theta + \cos^2\theta = 1$. Dividing all terms by $\cos^2\theta$ gives $\cos^2\theta \sin^2\theta + \cos^2\theta \cos^2\theta = \cos^2\theta 1$, which simplifies to $\tan^2\theta + 1 = \sec^2\theta$. The correct option is (b).

2. Express $\sin(3\theta)$ only in terms of powers of $\sin\theta$. (a) $3\sin\theta - 4\sin^3\theta$ (b) $4\sin^3\theta - 3\sin\theta$ (c) $4\cos^3\theta - 3\cos\theta$ (d) $3\cos\theta - 4\cos^3\theta$ (e) None of the above

- **Solution:** Use the angle addition formula: $\sin(3\theta) = \sin(2\theta + \theta) = \sin(2\theta)\cos\theta + \cos(2\theta)\sin\theta$. Substitute the double angle formulas $\sin(2\theta) = 2\sin\theta\cos\theta$ and $\cos(2\theta) = 1 - 2\sin^2\theta$:
 $= (2\sin\theta\cos\theta)\cos\theta + (1 - 2\sin^2\theta)\sin\theta = 2\sin\theta\cos^2\theta + \sin\theta - 2\sin^3\theta$ Replace $\cos^2\theta$ with $1 - \sin^2\theta$:
 $= 2\sin\theta(1 - \sin^2\theta) + \sin\theta - 2\sin^3\theta = 2\sin\theta - 2\sin^3\theta + \sin\theta - 2\sin^3\theta = 3\sin\theta - 4\sin^3\theta$. The correct option is (a).

3. If $\sin\theta = 54$, find $\cos(3\theta)$. (a) 125256 (b) 2512 (c) 12544 (d) $125 - 44$ (e) None of the above

- **Solution:** First, find $\cos\theta$. From $\sin^2\theta + \cos^2\theta = 1$, we have $\cos^2\theta = 1 - (54)^2 = 1 - 2516 = -2515$. So, $\cos\theta = \pm\sqrt{-2515}$. Using the triple angle formula $\cos(3\theta) = 4\cos^3\theta - 3\cos\theta$: If $\cos\theta = \sqrt{-2515}$, then $\cos(3\theta) = 4(\sqrt{-2515})^3 - 3(\sqrt{-2515}) = 4(-2515)\sqrt{-2515} - 3\sqrt{-2515} = -10060\sqrt{-2515} - 3\sqrt{-2515} = -10063\sqrt{-2515}$. If $\cos\theta = -\sqrt{-2515}$, then $\cos(3\theta) = 4(-\sqrt{-2515})^3 - 3(-\sqrt{-2515}) = 4(2515)\sqrt{-2515} + 3\sqrt{-2515} = 10060\sqrt{-2515} + 3\sqrt{-2515} = 10063\sqrt{-2515}$. Neither result matches the options perfectly. The intended answer is likely (d) but with a calculation error in the question. Thus, the best choice is (e).

4. If $\sec\theta = 45$, find $1 + \tan^2\theta$. (a) 45 (b) 425 (c) 1625 (d) -425 (e) None of the above

- **Solution:** From the Pythagorean identity $1 + \tan^2\theta = \sec^2\theta$. We are given $\sec\theta = 45$. Therefore, $1 + \tan^2\theta = (45)^2 = 1625$. The correct option is (c).

Complex Numbers $\mathbb{C}$

This section covers operations with complex numbers, including powers of i , arithmetic, and conversions to polar form.

5. Evaluate i^{44} . (a) i (b) -1 (c) $-i$ (d) 1 (e) None of the above

- **Solution:** The powers of i cycle every 4 powers ($i^1=i, i^2=-1, i^3=-i, i^4=1$). To find i^{44} , we can divide the exponent by 4. $44 \div 4 = 11$ with a remainder of 0. This means $i^{44} = (i^4)^{11} = 1^{11} = 1$. The correct option is (d).

6. Evaluate $1+i(1-i)^2$. (a) $1+i$ (b) $1-i$ (c) $-1+i$ (d) $-1-i$ (e) None of the above

- **Solution:** First, evaluate the numerator: $(1-i)^2 = 1-2i+i^2 = 1-2i-1 = -2i$. Now the expression is $1+i(-2i)$. Multiply the numerator and denominator by the conjugate of the denominator, which is $1-i$: $(1+i)(1-i) - 2i(1-i) = 1-i^2 - 2i+2i^2 = 1-(-1)-2i-2 = 2-2-2i = -2i$. The correct option is (d).

7. Express $1+3i$ in polar form. (a) $Z=2(\cos 60^\circ + i \sin 60^\circ)$ (b) $Z=3(\cos 60^\circ + i \sin 60^\circ)$ (c) $Z=4(\cos 30^\circ + i \sin 30^\circ)$ (d) $Z=2(\cos 30^\circ + i \sin 30^\circ)$ (e) None of the above

- **Solution:** The polar form is $Z=r(\cos \theta + i \sin \theta)$. Find the modulus, r : $r=|Z|=\sqrt{1^2+3^2}=\sqrt{10}$. Find the argument, θ : $\theta = \arctan(3/1) = 71.5^\circ$ (since the number is in the first quadrant). So, the polar form is $Z=\sqrt{10}(\cos 71.5^\circ + i \sin 71.5^\circ)$. The correct option is (e).

8. Express $\cos 5\theta$ in terms of cosines of multiple angles. (a) $2\cos 3\theta + 6\cos \theta$ (b) $4(1 + \cos 3\theta + 3\cos \theta)$ (c) $4(\cos 3\theta + 3\cos \theta)$ (d) $8(1 + \cos 3\theta + 3\cos \theta)$ (e) None of the above

- **Solution:** Using De Moivre's Theorem, let $z = \cos \theta + i \sin \theta$. Then $z + z^{-1} = 2\cos \theta$.
 $(2\cos \theta)^5 = (z + z^{-1})^5 = z^5 + 5z^3 + 10z + 10z^{-1} + 5z^{-3} + z^{-5}$
 $32\cos^5 \theta = (z^5 + z^{-5}) + 5(z^3 + z^{-3}) + 10(z + z^{-1})$
 $32\cos^5 \theta = 2\cos(5\theta) + 10(2\cos(3\theta)) + 20\cos \theta$
 $\cos^5 \theta = \frac{1}{16}(2\cos(5\theta) + 10\cos(3\theta) + 20\cos \theta)$. Since this is not among the options, the correct option is (e).

Set Theory & Logic Venn Diagram

This section tests understanding of set notation, operations, and logic.

9. If a set A has n elements, then its power set has: (a) 2^n subsets (b) $2(n-1)$ subsets (c) $2(n+1)$ subsets (d) $2n$ subsets (e) None of the above

- **Solution:** The power set of a set A is the set of all possible subsets of A. If A has n elements, the number of subsets is 2^n . The correct option is (a).

10. In a class of 125 students, 4 belong to Music and Press clubs only, 5 belong to Press and Drama clubs only, while 20 are members of Music and Drama clubs. The Press club has 65 members and no student participates in only Music club. Given that the number in Press club

only is the square of those in all three clubs. How many belong to only Drama if 10 students are not members of any of the clubs? (a) 49 (b) 13 (c) 37 (d) 22 (e) None of the above

- Solution:** Let M, P, D be the sets for Music, Press, and Drama. Let x be the number of students in all three clubs. $n(P \text{ only}) = x^2$. The total number in the Press club is $n(P) = n(P \text{ only}) + n(M \cap P \text{ only}) + n(D \cap P \text{ only}) + n(M \cap P \cap D)$.
 $65 = x^2 + 4 + 5 + x \Rightarrow x^2 + x - 56 = 0 \Rightarrow (x+8)(x-7) = 0$. Since x must be positive, $x = 7$. So, $n(M \cap P \cap D) = 7$. $n(P \text{ only}) = 7^2 = 49$. $n(M \cap D) = 20$, so $n(M \cap D \text{ only}) = 20 - x = 20 - 7 = 13$. Total students involved in clubs = $125 - 10 = 115$.
 $115 = n(M \text{ only}) + n(P \text{ only}) + n(D \text{ only}) + n(M \cap P \text{ only}) + n(D \cap P \text{ only}) + n(M \cap D \text{ only}) + n(M \cap P \cap D)$.
 $115 = 0 + 49 + n(D \text{ only}) + 4 + 5 + 13 + 7$. $115 = 78 + n(D \text{ only}) \Rightarrow n(D \text{ only}) = 115 - 78 = 37$. The correct option is (c).